**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 152 · Paper 1: CPU, Memory and Storage

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. For a numeric age input, give erroneous data of the wrong type.

Q2. What does a record contain?
A. Only one unnamed value
B. Only network addresses
C. Only one fixed data type
D. Named fields about one item

Q3. “Correcting one line proves the entire program is now fault-free without further testing.” Correct this. Explain why.

Q4. A program calculates an area and shows a menu in several places. Correct both claims. Explain why. (a) A function cannot return a value. (b) A procedure must return a value.

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# I Do · Read new knowledge

### Key vocabulary

CPU registers — PC, MAR, MDR and ACC have distinct jobs during instruction processing.

Primary memory — RAM is volatile working memory; ROM non-volatile firmware; cache accelerates access.

Storage selection — Compare SSD, HDD and optical media against contextual constraints.

Extended responses — Explain causal mechanisms, not merely 'faster' or 'cheaper'.

### New learning

CPU registers: PC, MAR, MDR and ACC have distinct jobs during instruction processing.

Primary memory: RAM is volatile working memory; ROM non-volatile firmware; cache accelerates access.

Storage selection: Compare SSD, HDD and optical media against contextual constraints.

Extended responses: Explain causal mechanisms, not merely 'faster' or 'cheaper'.

### Worked example

Explain how cache can help a CPU running a frequently repeated program.

1. Repeated instructions/data may be present in cache.

2. Cache access is typically faster than RAM.

3. CPU may avoid slower RAM fetches on a cache hit.

4. This can improve execution performance.

# We Do · Practise with guidance

### Guided 1

A mobile field worker needs portable storage. Explain one benefit and drawback of SSD.

Worked method: Give scenario-linked balanced conclusion.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Predict the final result after two assignments to a working variable.

```
acc = 6
acc = acc + 5
print(acc)
```

predict

### Question 1

What does MDR hold during a memory operation? [2 marks]

### Question 2

Give one difference between RAM and ROM. [2 marks]

### Question 3

Explain one disadvantage of an HDD in a moving laptop. [3 marks]

## Exit Ticket · Answer before checking

ROM is the volatile store for currently running applications.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. The text 'twelve' is the wrong data type and must be handled or rejected.

Q2. D — Named fields about one item

Q3. To check that the fault is fixed and that other required behaviour still works.

Q4. (a) A function returns a value, such as an area. (b) A procedure performs a task, such as showing a menu, without returning a value. Use arguments and return values to avoid unwanted changes to global data.

### I Do / We Do pages · Guided practice

Q1. Give scenario-linked balanced conclusion.

### You Do pages · Code First

```
acc = 6
acc = acc + 5
print(acc)
```

Expected result: 11

### You Do pages · Independent questions

Q1. Data or instruction being transferred.

Q2. RAM is volatile working memory; ROM is non-volatile and commonly stores firmware.

Q3. Moving mechanical parts make it more vulnerable to physical shock.

### Exit Ticket · Model answer

False

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## 2. Write a complete new answer in GREEN PEN. Show working if needed:

| Do Now | I Do | We Do | You Do | Review & Improve |
|---|---|---|---|---|
| Recall previous learning | Read new knowledge | Practise with guidance | Work independently | Check understanding |